Slide 1: Hotspot Detection & Early Exit Strategy

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Objectives & Challenges

Goal

Terminate immediately upon finding a row with **no local hotspots**.

Challenge

Standard C `break` is illegal in OpenMP loops. We must break a parallel loop barrier instantly without leaving zombie threads.

Solution: OpenMP Cancellation

`#pragma omp cancel for` `#pragma omp cancellation point for`

Synchronization

Used `#pragma omp critical` to safely set `early_exit` and `bad_row`, ensuring only the first offending row is reported.

Runtime Requirement

OpenMP disables cancellation by default for performance. We must set `export OMP_CANCELLATION=true` before execution.



heatmap_analysis_quick.c

```c
int early_exit = 0;  bad_row = -1;

#pragma omp parallel
{
    #pragma omp for schedule(static)
    for (int i = 0; i < rows; ++i) {
        #pragma omp cancellation point for

        int row_hotspots = 0;
        /* ... count local hotspots in row i ... */

        if (row_hotspots == 0) {
            #pragma omp critical
            {
                if (!early_exit) {
                    early_exit = 1;  bad_row = i;
                    printf("Row %d contains no hotspots.\n", i);
                    printf("Early exit.\n");
                }
            }
            #pragma omp cancel for
        }
    }
}
```